

⇒ Random Forest → reduce variance

from

$$M = \text{Agg}(M_1, M_2, M_3, \dots, M_K)$$

overfit / under

Cross R → 

RvsR → 

$K \uparrow$ variance $\downarrow$

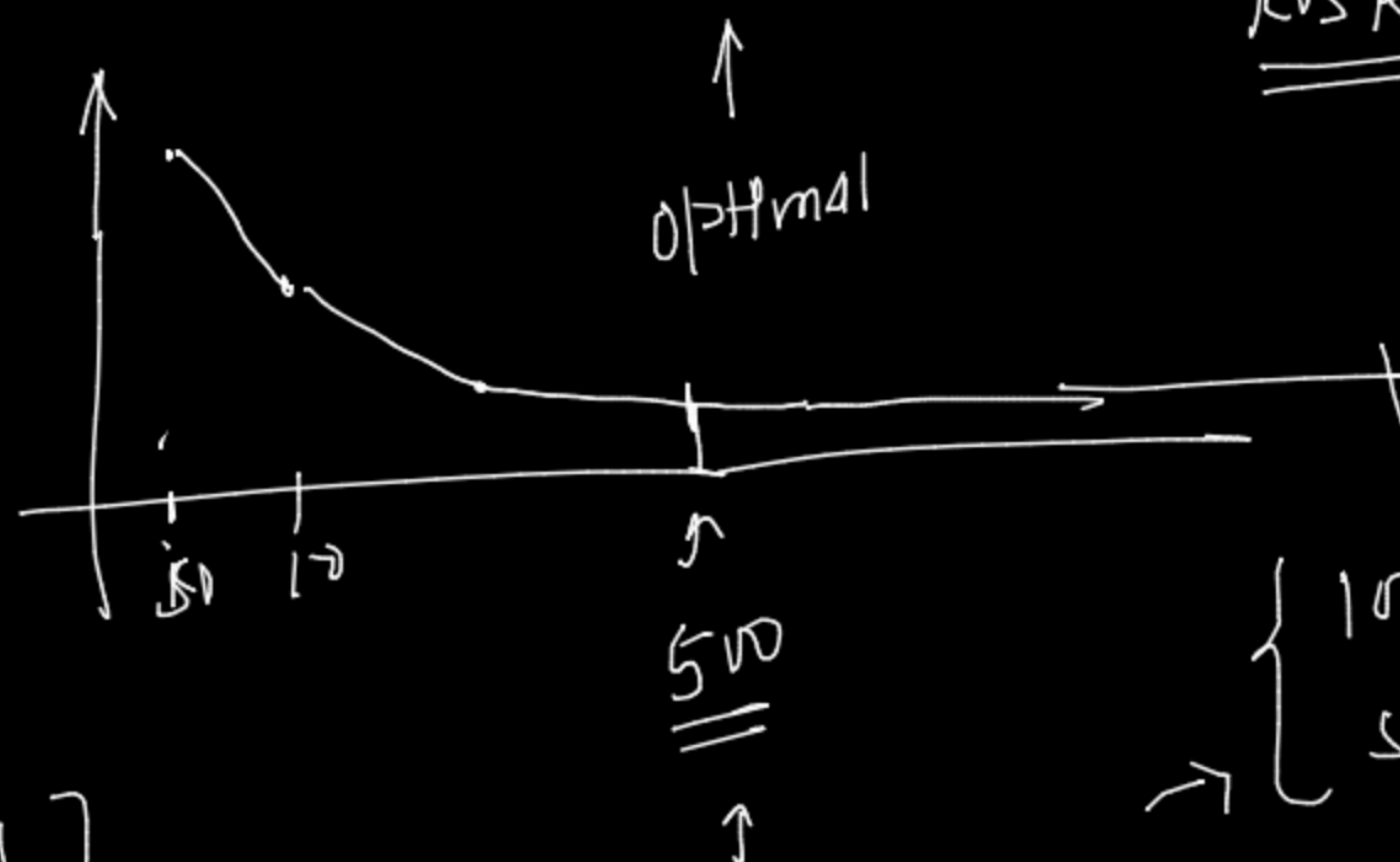
$K \downarrow$ variance $\uparrow$

$K=1$ $K=$

DT

documentation

n-estimators $\rightarrow$ $\left. \begin{matrix} (100) \\ 20 \\ 10 \end{matrix} \right\}$
 $\rightarrow$ $\textcircled{RF}$



$\left. \begin{matrix} 1000 \\ 500 \end{matrix} \right\}$

Boosting →

Talking: Rahul Tripathi

low variance + high biased

+ Randomisation

+ Aggregation

(LR)

(RS + CS)

(error ↓)

idea (i)

Result

⇒

low variance + low biased model



Core Idea

$$D_n = \{x_i, y_i\}_{i=1}^n$$

stage 0
CS & R-S

$$D_{train} = \{x_i, y_i\}_{i=1}^n$$

→ M_0

↳ high bias
low variance

$$y = h_0(x) \Rightarrow 270$$

↑
210

error

$$y_i - h_0(x_i)$$

X	Y	$\hat{y}_0$	error	$\hat{y}$	error
3	250	270	-20	-10	-20
4	270	210	-60	-50	-10

→ ...

↓ Pass to Next stage

at stage 1

$$\{x_i, y_i, error_i\}$$

$$\{x_i, error_i\} \leftarrow M_1$$

↑

$$\hat{y} = h_1(x)$$

$$\boxed{M_1} \leftarrow \{x_i; \text{error}_i\}$$

$\Downarrow$

$$h_1(x) \quad \underline{\text{error}} \approx y_i - h_0(x_i)$$

$\uparrow$

3/4

at state 0

state

$$\Rightarrow F_1(x) = \alpha_0 h_0(x) + \alpha_1 h_1(x)$$

$\Downarrow$

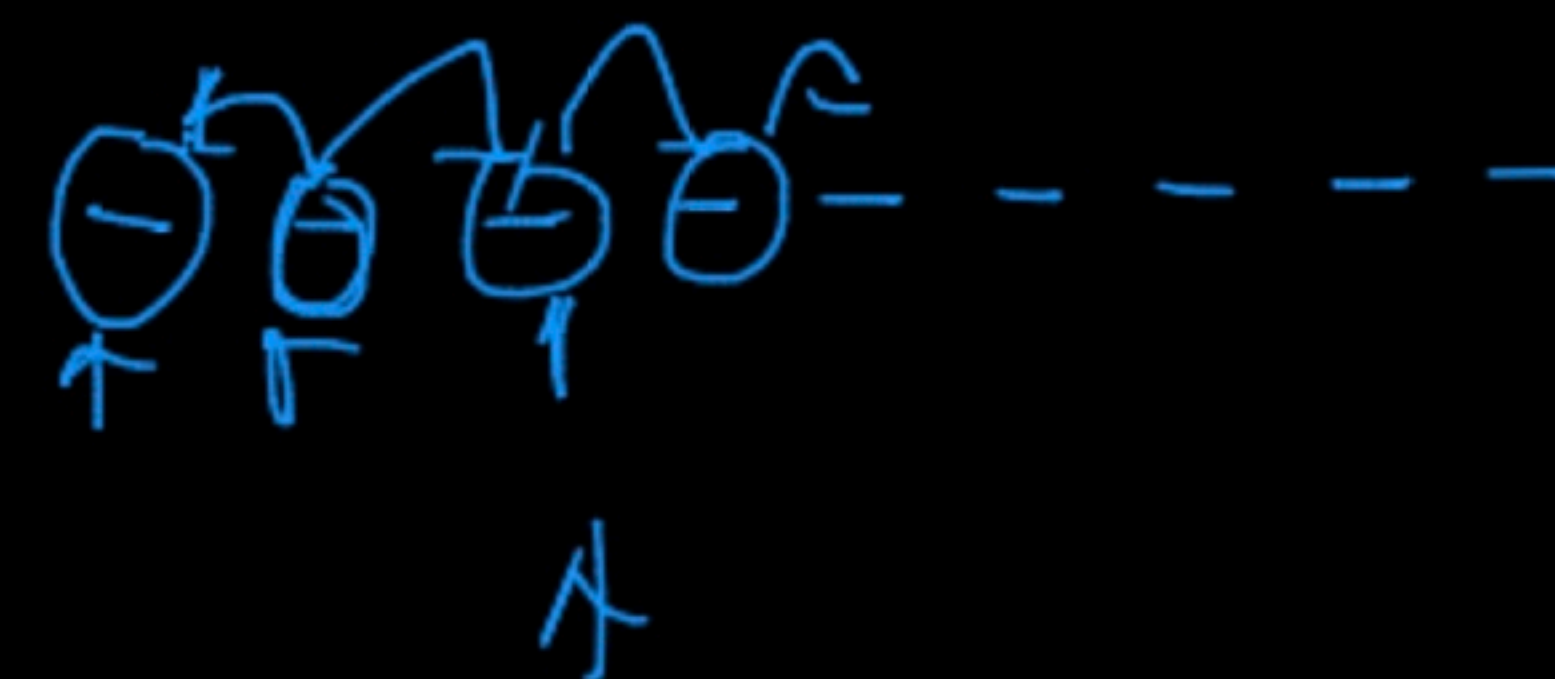
at 2nd state

$\{x_i; \text{error}_i\}$

$$\rightarrow (y_i - F_1(x_i))$$

error $\downarrow$

$\left. \begin{array}{c} - \\ - \\ - \end{array} \right\} \rightarrow$



Repeat till convergence